

Design Brief

Architectural Decisions

This service is built on a **decoupled, asynchronous architecture** to ensure high availability and responsiveness:

- **API Orchestration:** FastAPI serves as the entry point, providing a non-blocking interface for file uploads and metadata retrieval.
- **Async Task Processing:** Heavy AI workloads (STT, Diarization, Sentiment Analysis) are offloaded to **Celery workers** via a **Redis** broker. This prevents API timeouts and allows for reliable background processing.
- **Relational Persistence:** **PostgreSQL** handles structured data, enabling complex queries for "coachable moments" across thousands of call segments.
- **Containerized Isolation:** Using **Docker Compose**, each component (API, Worker, DB, Cache) is isolated, ensuring a reproducible environment and simplified dependency management.

Horizontal Scalability

The system is designed to grow with demand by scaling components independently:

- **Independent Scaling:** API pods can scale to handle request traffic, while Worker nodes scale based on **Queue Depth** (transcription backlog).
- **Compute vs. I/O:** AI-heavy workers can be assigned to high-memory instances without affecting the lightweight API layer.
- **Production Storage:** While the MVP uses Docker volumes, the architecture is ready to switch to **Object Storage (S3/GCS)** for multi-node file access.

Fault-Tolerance & Reliability

- **Task Retries:** Celery is configured with **exponential backoff** for transient failures (e.g., Deepgram API blips or network issues).
- **Process Isolation:** A failure in the STT worker does not impact the API or TTS generation, preventing single points of failure.
- **Resource Management:** Models are loaded once during **Worker Warm-up** to prevent memory leaks and ensure consistent latency.

Trade-offs & Considerations

- **Managed APIs vs. Self-Hosting:** Chose **Deepgram** over WhisperX for the MVP to ensure a "zero-friction" setup for reviewers (avoiding Hugging Face gated model tokens).
- **Storage Locality:** Local volumes provide speed for 30-60s snippets but would be transitioned to S3 for true horizontal scaling.
- **Image Weight:** The sentiment model is pre-baked into the image to eliminate startup latency, accepting a larger image size for better **runtime performance**.